

Pascal's Row Sum

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1 PascalRowSum

Prove that:

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} = 2^n$$

Draw Pascal's Triangle.

$$\binom{0}{0}$$

$$\binom{1}{0} \quad \binom{1}{1}$$

$$\binom{2}{0} \quad \binom{2}{1} \quad \binom{2}{2}$$

$$\binom{3}{0} \quad \binom{3}{1} \quad \binom{3}{2} \quad \binom{3}{3}$$

$$\binom{4}{0} \quad \binom{4}{1} \quad \binom{4}{2} \quad \binom{4}{3} \quad \binom{4}{4}$$

Notice, the sum of the n th row seems to be 2^n .

1.0.1 Sidenote 1

Prove that

$$\binom{p}{0} = \binom{p+1}{0}$$

Proof:

$$\binom{p}{0} = \frac{p!}{0! \cdot (p-0)!} = \frac{p!}{p!} = 1 = \frac{(p+1)!}{0! \cdot (p+1-0)!} = \binom{p+1}{0}$$

1.0.2 Sidenote 2

Prove that

$$\binom{p}{p} = \binom{p+1}{p+1}$$

Proof:

$$\binom{p}{p} = \frac{p!}{0! \cdot (p-0)!} = \frac{p!}{p!} = 1 = \frac{(p+1)!}{0! \cdot (p+1-0)!} = \binom{p+1}{p+1}$$

1.0.3 Sidenote 3

Prove that

$$\binom{p}{q} + \binom{p}{q+1} = \binom{p+1}{q+1}$$

Proof:

$$\begin{aligned} \binom{p}{q} + \binom{p}{q+1} &= \frac{p!}{q! \cdot (p-q)!} + \frac{p!}{(q+1)! \cdot (p-q-1)!} \\ &= \frac{p! \cdot (q+1)}{(q+1)! \cdot (p-q)!} + \frac{p! \cdot (p-q)}{(q+1)! \cdot (p-q)!} \\ &= \frac{p! \cdot (q+1+p-q)}{(q+1)! \cdot (p-q)!} = \frac{p! \cdot (p+1)}{(q+1)! \cdot (p-q)!} \\ &= \frac{(p+1)!}{(q+1)! \cdot (p-q)!} = \binom{p+1}{q+1} \end{aligned}$$

Using induction:

for n=0:

$$\binom{0}{0} = \frac{0!}{0! \cdot 0!} = 1 = 2^0$$

Assume n=k works:

$$\binom{k}{0} + \binom{k}{1} + \binom{k}{2} + \dots + \binom{k}{k-2} + \binom{k}{k-1} + \binom{k}{k} = 2^k$$

Prove for n=k+1

$$\binom{k+1}{0} + \binom{k+1}{1} + \binom{k+1}{2} + \dots + \binom{k+1}{k-1} + \binom{k+1}{k} + \binom{k+1}{k+1}$$

Using sidenote 1, replace $\binom{k+1}{0}$ with $\binom{k}{0}$

Using sidenote 2, replace $\binom{k+1}{k+1}$ with $\binom{k}{k}$

Using sidenote 2, replace $\binom{k+1}{q+1}$ with $\binom{k}{q} + \binom{k}{q+1}$

$$\begin{aligned} &= \binom{k}{0} + \binom{k}{0} + \binom{k}{1} + \binom{k}{1} + \binom{k}{2} + \dots + \binom{k}{k-2} + \binom{k}{k-1} + \binom{k}{k-1} + \binom{k}{k} + \binom{k}{k} \\ &= 2 \cdot \left(\binom{k}{0} + \binom{k}{1} + \binom{k}{2} + \dots + \binom{k}{k-2} + \binom{k}{k-1} + \binom{k}{k} \right) = 2 \cdot (2^k) = 2^{k+1} \end{aligned}$$

Therefore, proven by induction.